MILITARY SPECIFICATION

SOCKS, FLYER'S, ANTIEXPOSURE CWU-67/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of flyer's antiexposure socks, that are designated CWU-67/P.

1.2 Classification. The socks shall be of one type in the following sizes, as specified (see 6.2).

Medium (M)

Large (L)

2. APPLICABLE DOCUMENTS

2.1 *Government documents*

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

U-T-30	Talcum Powder
V-T-295	Thread, Nylon
UU-P-268	Paper, Kraft, Wrapping
DDD-L-20	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-A-5540	Adhesive, Polychloroprene
MIL-C-23926	Cloth, Coated and Laminated, Polychloroprene on Nylon, and Tape, Polychloroprene, Unsupported

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads

2.1.2 Other Government drawings. The following Government drawing forms a part of this specification to the extent specified herein.

DRAWING

AIR FORCE

78D139	Patterns, Socks, Flyer's, Antiexposure CWU-67/P

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. The issues of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard samples. Standard samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified (see 4.2, 4.3, and 6.2), the contractor shall furnish samples for first article inspection and approval.

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3.3 Material

3.3.1 Basic material. The socks shall be fabricated from a knitted stretch nylon uncured coated cloth conforming to MIL-C-23926, type IV. The color of the cloth shall be Sage Green, USAF Color Shade 1509.

3.3.2 Outersole. The outersole shall be fabricated from a knitted stretch nylon coated cloth conforming to MIL-C-23926, type I. The color of the cloth shall be Sage Green, USAF Color Shade 1509.

3.3.3 Tape. The unsupported tape for covering the sewn seams shall conform to MIL-C-23926, type III, class 1 or 2. The width of the unsupported tape, for covering the stitching, shall be 3/4 ±1/16 inch wide. The color shall be Sage Green, USAF Color Shade 1509.

3.3.4 Thread. The thread shall conform to type I or II, class 1 or 2, size E of V-T-295 and shall approximately match the color of the basic material. The dyed thread shall have at least good colorfastness to light and laundering.

3.3.5 Adhesive. The adhesive, for the cementing operation, shall conform to MIL-A-5540, class 1 or 2. The adhesive shall be neutral in color or colorless.

3.3.6 Dusting. After the seams have been taped and have thoroughly dried, the seams shall be dusted with talcum powder conforming to U-T-30.

3.3.7 Combination label. The inner side of each sock shall have a combination identification-size label conforming to type III, classes 1 and 2 of DDD-L-20 with 1/8 inch lettering. The lettering shall start 1 inch from the top of the socks and shall contain the following information:

Socks, Flyer's, Antiexposure CWU-67/P
Size: L (Example)
Stock No.: 8475-00-000-0000 (Example)
Supplier's Name:

3.4 Design. The sock shall be of one piece with an attached outersole.

3.5 Patterns. Standard patterns, which provide proper allowance for all seams, will be furnished by the Government. The Government patterns, which show directional lines and markings for proper assembly, shall not be altered in any way and are to be used as a guide for cutting supplier's working patterns. The working patterns shall be identical to the Government patterns except that additional notching to facilitate manufacture is permissable.

3.5.1 Pattern parts. The components of the socks shall be cut from the material, as specified below, in accordance with the pattern parts indicated:

Material	Pattern nomenclature	Cut parts
Basic material (see 3.3.1)	Sock	2
Outersole (see 3.3.2)	Sock outersole	2

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3.6 Construction. The construction of the socks shall be accomplished by use of sewn and cemented taped seams, as applicable, and the details of the construction shall conform to figures 1 and 2. The coated side of the sock shall be on the outer side of the sock.

3.6.1 Socks. The socks shall be fabricated from the uncured coated cloth (see 3.3.1) and all the seams shall be on the inner (cloth) side of the socks, as specified in figure 1, View A, Sections B and C. The stitching shall be 1/8 ±1/32 inch from the cloth edge. The sock shall be placed on the left and right sock forms, as applicable, conforming to figure 2, with the uncured coating on the outerside. The sock shall be drawn tightly against the form to eliminate wrinkles and creases. The construction seam shall be covered on the outerside, from around the toe to the top of the sock with the unsupported tape (see 3.3.3) as specified in figure 1, View A. The unsupported tape shall be cemented to the bottom of the sock. The outersole shall extend up beyond the heel. The sock with the sole and tape attached and on the form shall be cured in accordance with the coating supplier's instructions.

3.6.2 Stitches, seams, and stitchings. Stitches, seams, and stitching used in the construction of the socks shall conform to FED-STD-751. The seam used in the fabrication of the socks shall conform to figure 1. The stitching used in the socks shall be accomplished with stitch type 301 and shall contain 8 to 10 stitches per inch. Skipped stitches and runoffs shall be repaired using stitch type 301. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch. The ends of continuous lines of stitchings that are not caught in other seams or stitchings shall overlap not less than 1/2 inch.

3.6.3 Cementing operations. The seam tape shall be cemented on the outside of the sock to cover the joining seam that perforates the sock. The stitched seam of the sock shall not be cleaned with solvent. The surface of the sock to which the cement is to be applied shall be thoroughly cleaned with a suitable solvent so that foreign matter including any dusting materials such as zinc stearate or talcum powder will be removed to prevent contamination of the cement. Brushing with a stiff bristled brush will be permitted. Care shall be exercised to prevent prolonged exposure of the basic fabric, the outersole fabric, and the thread to the solvent. The solvent shall have evaporated completely prior to the application of the cement and shall not leave any residue. All areas that are to be cemented shall be secured with the cement specified in 3.3.5 and in accordance with the recommendations of the manufacturer of the cement to achieve the optimum bonding properties and an adhesive of not less than 5 pounds. Not less than two coats of cement shall be used on all areas that are to be cemented. All seam tape shall be applied without tension. The cement shall be not more than 1/8 inch beyond the edges of the seam tape. Old cement or cement that has partly or completely polymerized shall not be used. Before containers are filled with fresh cement, the containers shall be freed of congealed cement. A fresh batch of cement shall be used at least every 8 hours during the manufacture of the socks.

3.6.4 Pairing. The socks, right and left, shall be matched in size and paired.

3.7 Workmanship. The finished socks shall conform to the quality of product established by this document. The finished socks shall be clean and free from any defects that might affect appearance or functionability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Components and materials inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of socks. The sample unit shall be one sock and the selection shall be by pairs. Defects for pairing shall be classified as a single defect. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot.

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions (see figure 2). Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of socks. The sample unit shall be one sock and the selection shall be by pairs. The inspection level shall be S-3 and the AQL, expressed in terms of defects per 100 units, shall be 4.0.

4.4.4 Critical defect examination. Every sock in lots found acceptable under the visual and dimensional examinations listed in 4.4.2 and 4.4.3 shall be examined for critical defects listed in 4.4.2. Any sock found to contain one or more critical defects shall be rejected.

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4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects sahll be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per 100 units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Open and noncontinuous heat sealed seams and closure of polyethylene bag. Omission of vent hole in polyethylene bag. Bulged or distorted container. Alternative closure of polyethylene bag not provided with tuck or reverse flap.
Contents	Number of intermediate packs per shipping container is more or less than required. Number of socks per intermediate pack is more or less than required. 1/ Size shown on one or more socks not as specified on shipping container. 1/

1/ For this defect, one intermediate pack from each container in the sample shall be examined.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The socks shall be inserted into a snug-fitting, clear polyethylene film bag of 0.00125 inch thickness (±25 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. A 1/4-inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Commercial. The socks shall be preserved in accordance with ASTM D 3951.

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TABLE I. Visual defects.

Defects	Classification		
	Critical	Major	Minor
Material defects and workmanship damages			
Hole, scissor or knife cut, tear, burn, abraded area, oil or soil mark, or any other weakening defects resulting from manufacturing process.	X		
Any spot or stain such as solvents, cements, or powders that cannot be removed without damage to fabric or components.	X		
Components and assembly			
Any defective component or defect that will seriously affect life saving characteristics of socks (unless otherwise classified herein).	X		
Any part not cut in accordance with pattern.		X	
Any component not specified color.			X
Cementing operations			
Cement on cloth surfaces along seam tape or attachment edges more than 1/8 inch.			X
Location of any seam tape not as specified.		X	
Any seam tape applied in crooked manner for distance exceeding 1 inch but not exceeding 2 inches in length (less than 1 inch in length not to be classified as defect).			X
Any seam tape applied in crooked manner for distance exceeding 2 inches in length.		X	
Any damaged seam tape edges greater than 1/16 inch from outside edge.			X
Seams and stitchings			
Thread ends not trimmed or loose thread ends not removed.			X
Any seam or attachment of any component twisted, puckered, pleated, or caught in any unrelated operation or stitching.		X	
Ends of stitching that are not caught in other seams or stitchings not overlapped at least 1/2 inch.		X	

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TABLE I. Visual defects. - Continued

Defects	Classification		
	Critical	Major	Minor
Thread breaks not secured by stitching back of break not less than 1/2 inch.			X
Skipped stitches or run off not repaired as specified herein.			X
Any stitching irregular or unevenly gaged on major portion of seam.			X
Seam or stitch type not as specified.		X	
Loose tension resulting in loose seam.		X	
Tight tension resulting in stitches breaking when normal pull is applied. 1/		X	
Stitches per inch on major portion of seam are one or two stitches per inch more or less than the specified stitches per inch.			X
Stitches per inch on major portion of seam are more than two stitches per inch more or less than the specified stitches per inch.		X	
Pairing			
Not properly mated, i.e., right and left socks not of the same size.		X	
Labeling			
Label missing, incorrect, incomplete, or illegible.			X

1/ Puckering is evidence of tight tension. If puckering is evident, normal pull shall be exerted in the lengthwise direction of the seam or stitching.

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5.2 Packing. Packing shall be level A, B, or Commerical as specified (see 6.2).

5.2.1 Level A. Twenty-four pairs of socks, of one size only, preserved as specified in 5.1, shall be packed in a weather resistant fiberboard shipping container conforming to PPP-B-636 style and grade optional. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15C of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each fiberboard container shall be closed in accordance with Method III, waterproofed in accordance with Method V, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Twenty-four pairs of socks, of one size only, preserved as specified in 5.1, shall be packed in a domestic fiberboard shipping container conforming to PPP-B-636, style and grade optional. The inside of each shipping container shall be fitted with a box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial. Socks preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Polyethylene packages. Each polyethylene bag (see 5.1.1) shall have the stock number, nomenclature, size, and quantity of socks contained therein legibly printed or stamped in black across the center face of the bag, or a white paper label with the required information printed thereon in black shall be placed in the bag so that it is readable through the polyethylene.

5.3.2 Mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label, 5 by 4 inches, with the words "MIXED NSNs" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and NSNs contained therein.

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5.4 Palletization. When specified (see 6.2), antiexposure socks, packed as specified in 5.2, shall be palletized in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversin the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. NOTES

6.1 Intended use. The socks covered by this specification are intended to b custom fitted to the individual and to be worn with MIL-C-83195, Coveralls, Flying, Anti-Exposure CWU-21/P.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. When a first article is required (see 3.2, 4.2 and 6.3).

d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

f. When palletization is required (see 5.4).

6.3 First article. First article samples shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The contracting officer should include specific instructions in all acquisition documents, regarding arrangements for inspection and approval of the first article samples.

6.4 Samples. For access to samples of the end item and shade samples, addres the contracting activity issuing the invitation for bids.

6.5 Figures. Figures 1 and 2 show general style of the socks and are furnished for information only. Should any inconsistencies exist between the requirements of this specification and the figures, the requirements of this specification shall govern.

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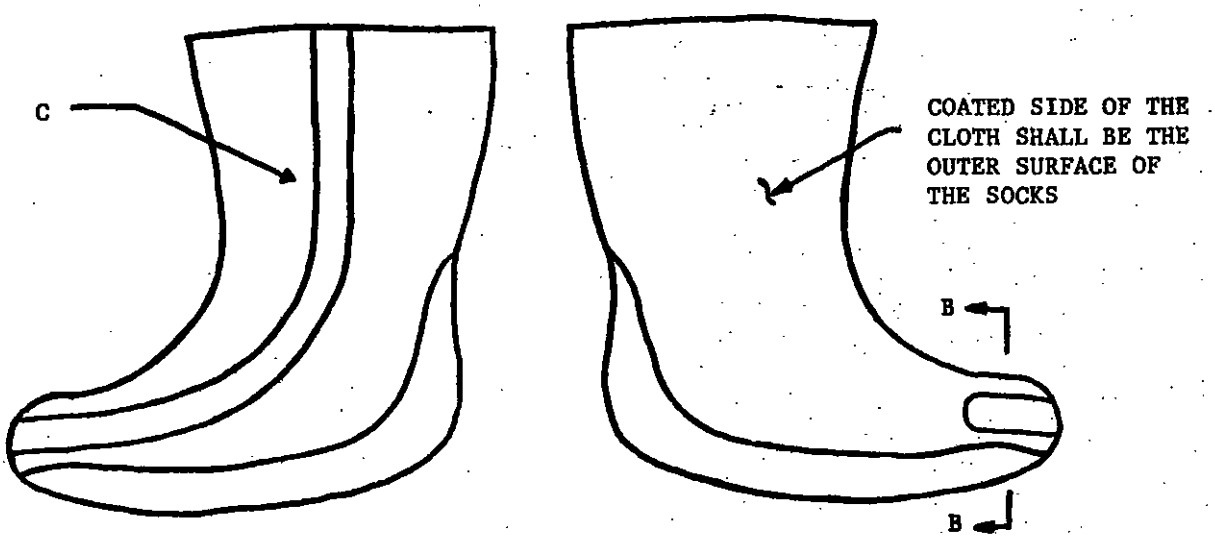
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Review activity:
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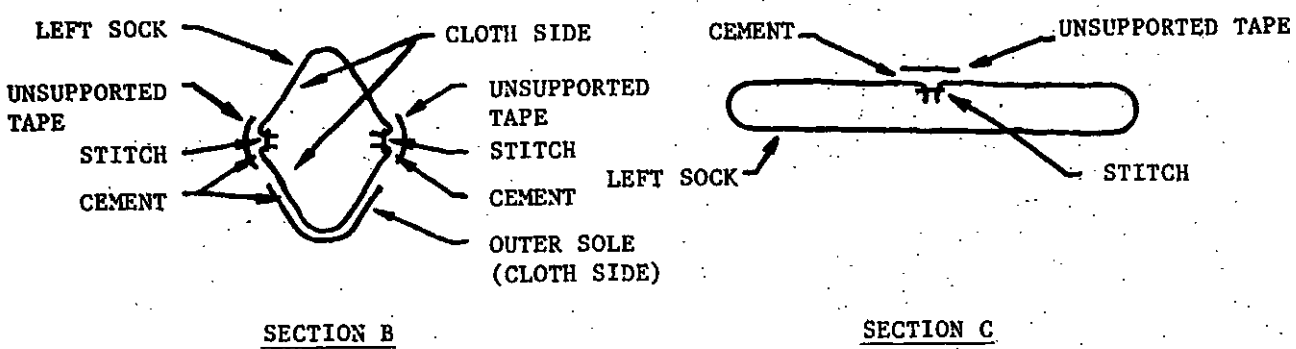
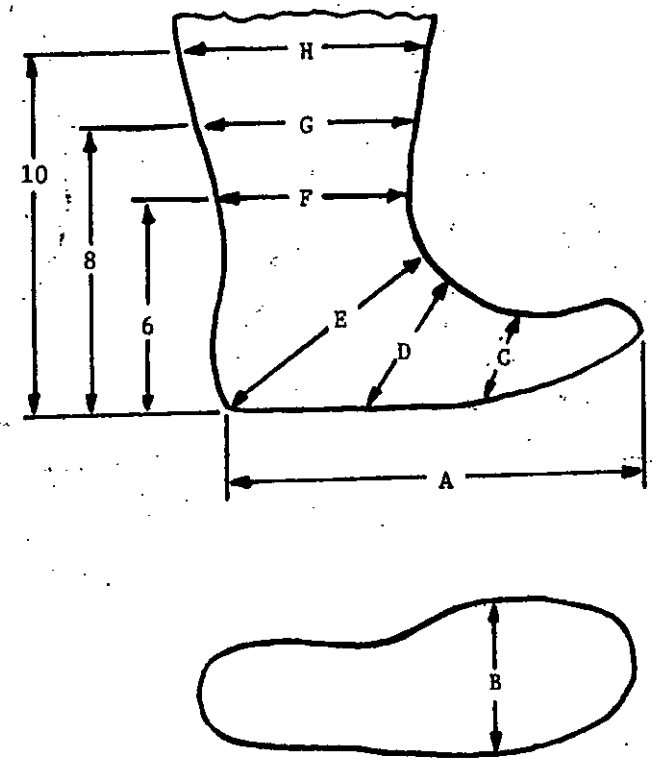


FIGURE 1. Left sock sections.

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DIMENSION	SIZE	
	MEDIUM	LARGE
A	11	11-11/16
B	3-9/16	3-3/4
C	9	9-1/2
D	9-3/4	10-1/4
E	15	15-3/4
F	12-3/4	13-1/4
G	15	15-1/4
H	17-5/8	17-5/8

NOTES:
1. Dimensions are in inches
2. Unless otherwise specified, tolerances are:

DIMENSIONS	TOLERANCES
0 to 1/2 incl.	±1/32
over 1/2 to 1 incl.	±1/16
over 1 to 6 incl.	±1/8
over 6 to 20 incl.	±1/4

FIGURE 2. Sock form.

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